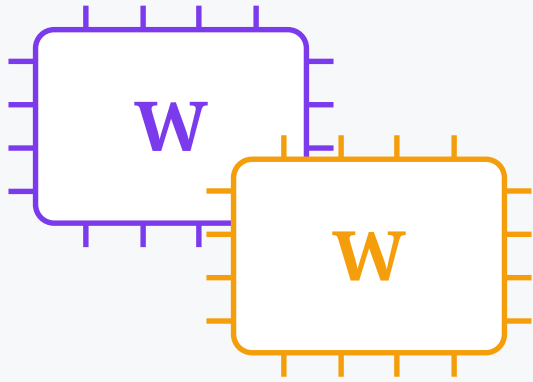




Hardware Model Checking Certification

with Certifaiger and Cerbtora

IJCAR 2026

Nils Froleys

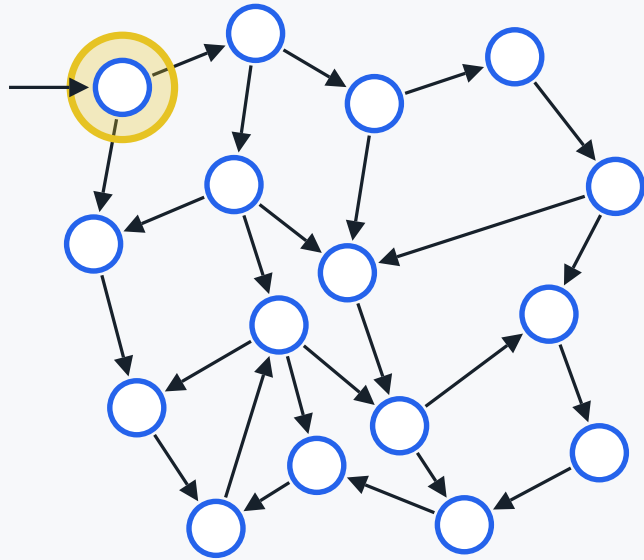
Emily Yu

KU LEUVEN



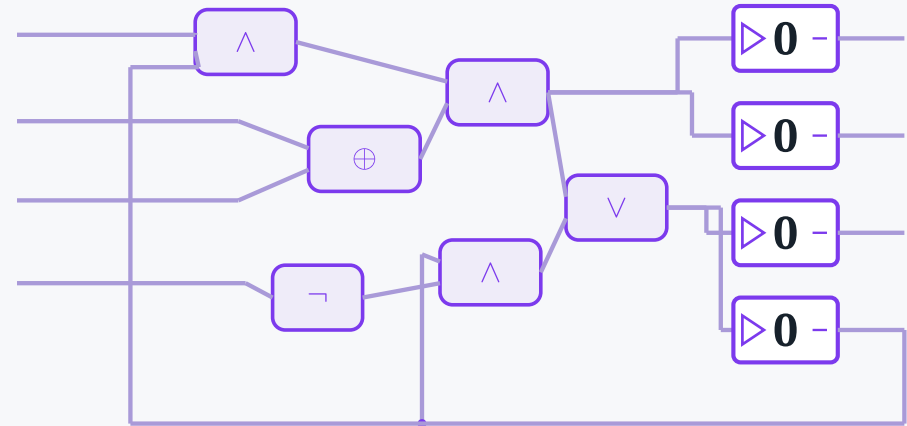
Universiteit
Leiden

Logic Circuits

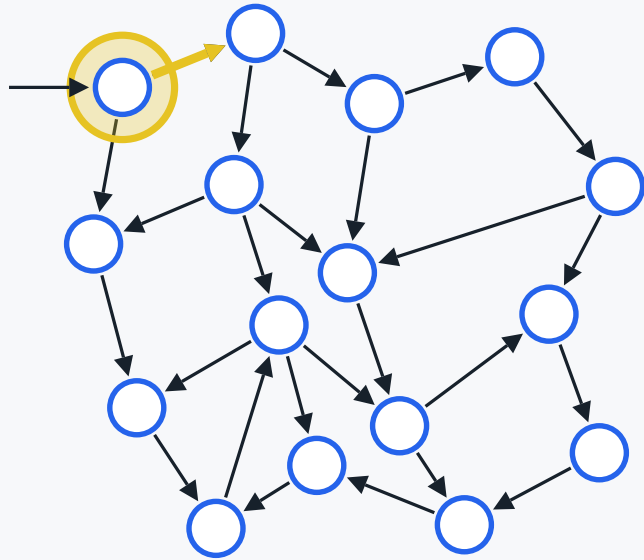


Bit Level

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P} \rangle$$

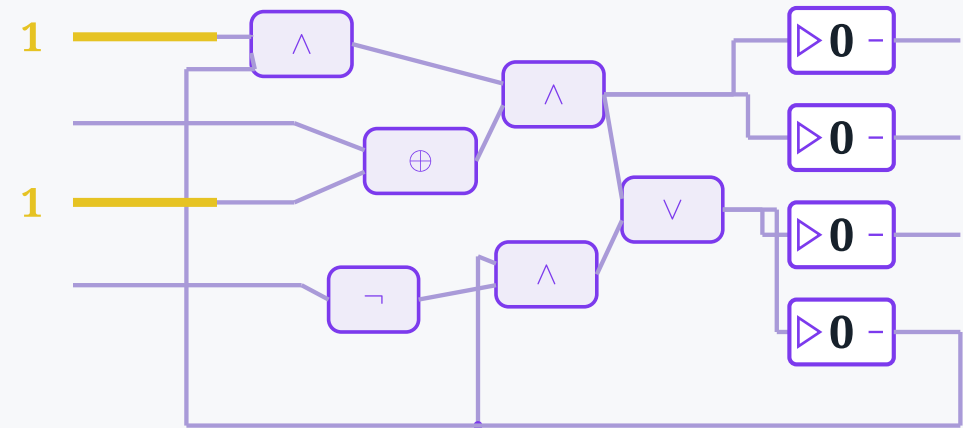


Logic Circuits

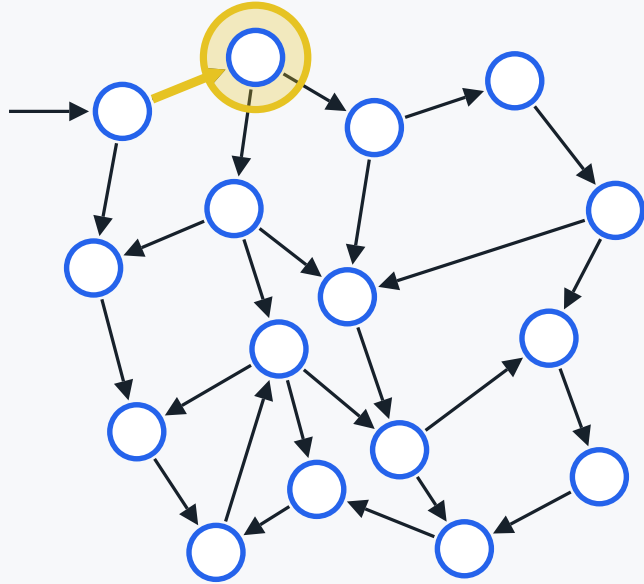


Bit Level

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P} \rangle$$

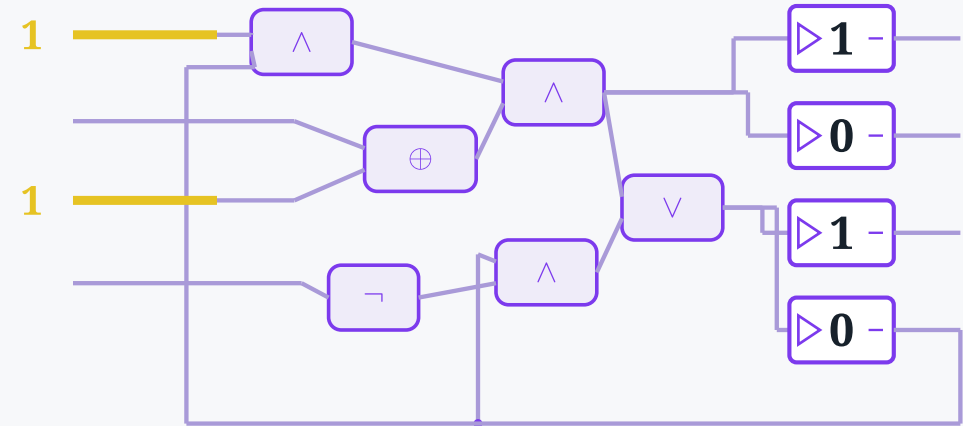


Logic Circuits

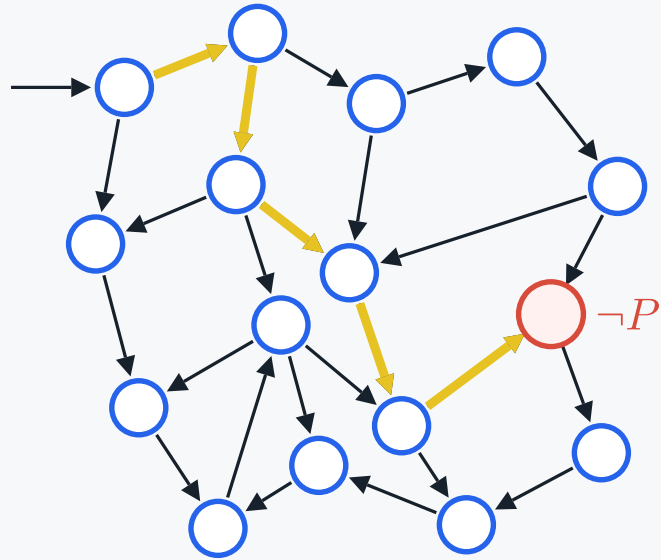


Bit Level

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P} \rangle$$

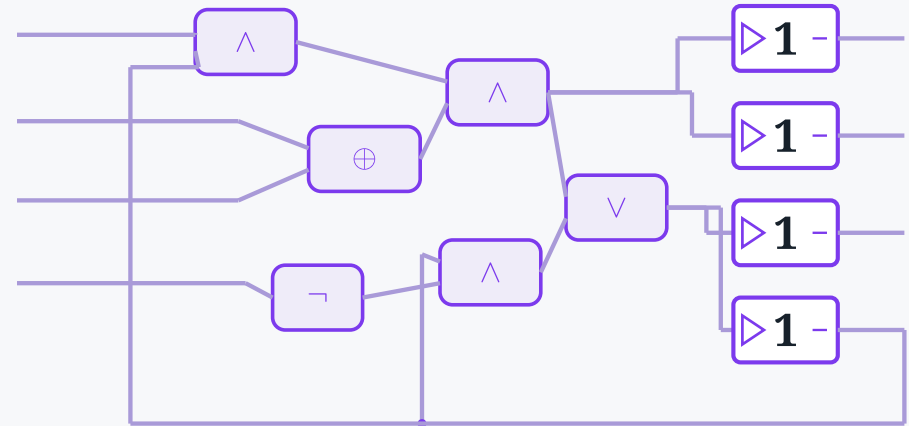


Logic Circuits

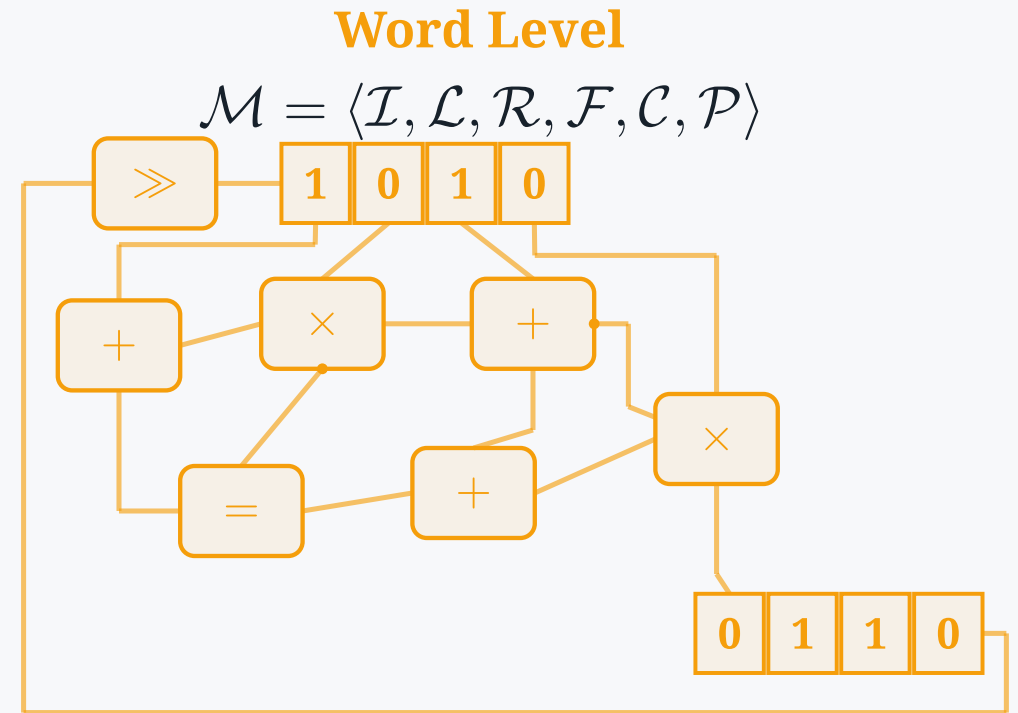
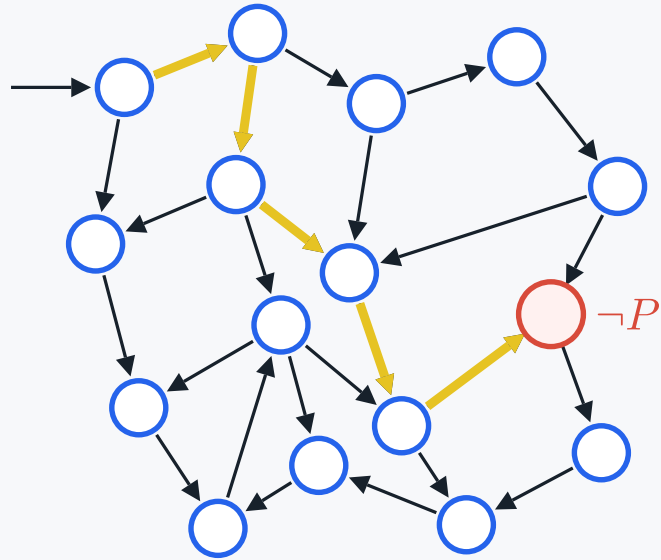


Bit Level

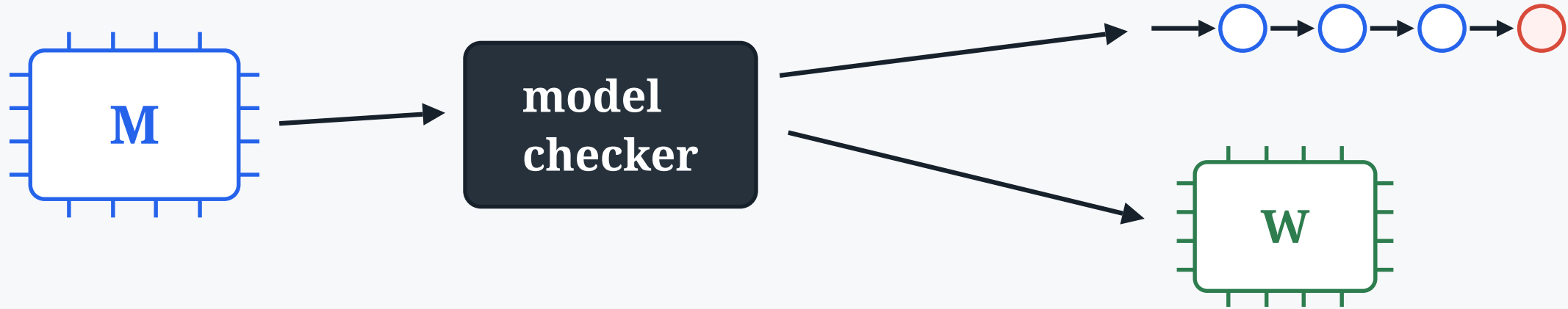
$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P} \rangle$$



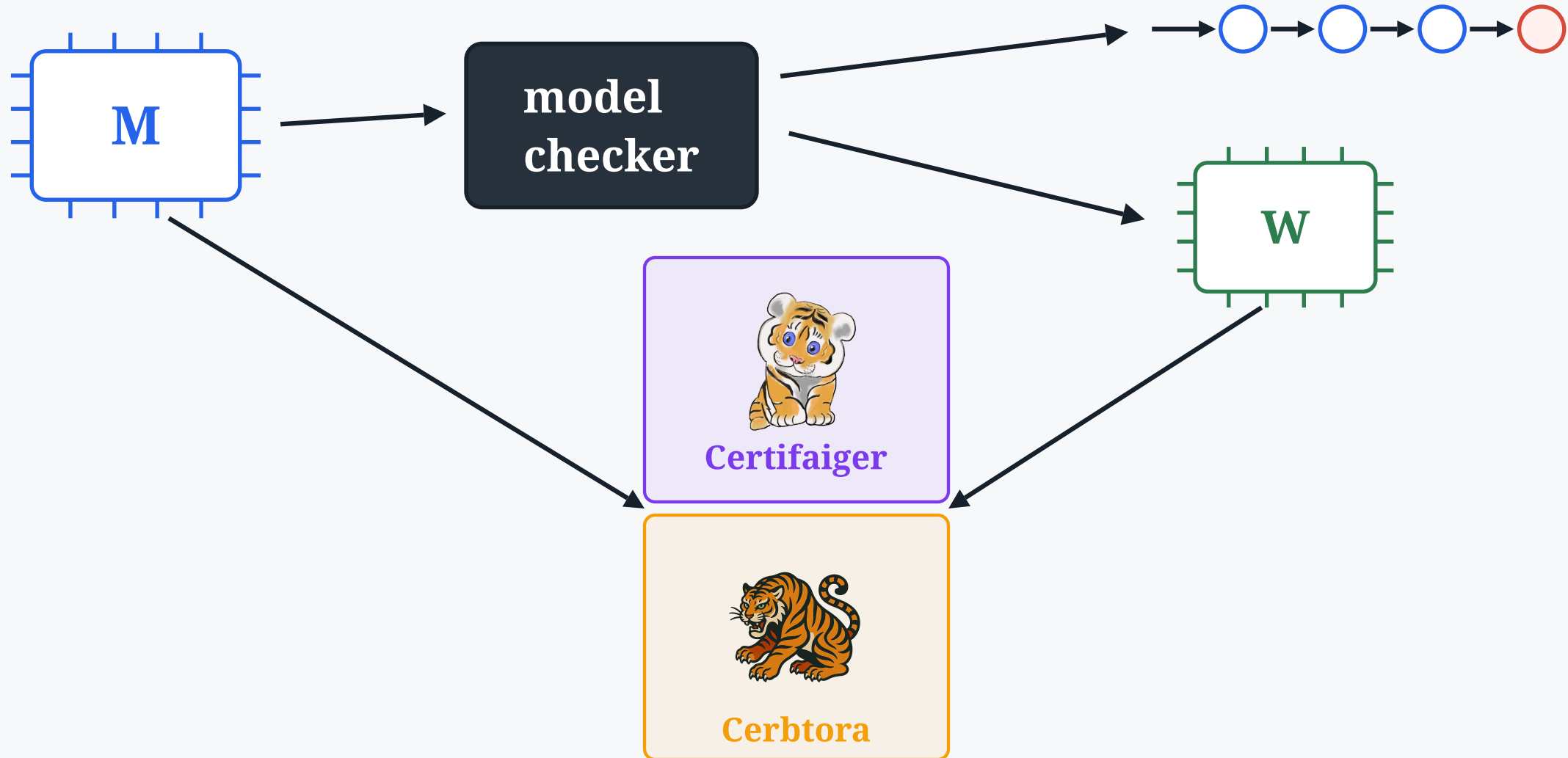
Logic Circuits



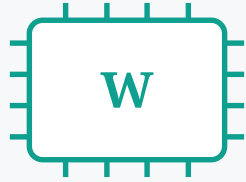
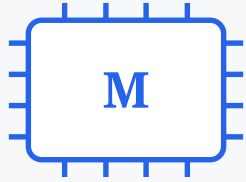
Certificate Checking



Certificate Checking



Certificate Conditions



$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P} \rangle \quad \mathcal{W} = \langle \mathcal{I}', \mathcal{L}', \mathcal{R}', \mathcal{F}', \mathcal{C}', \mathcal{P}' \rangle$$

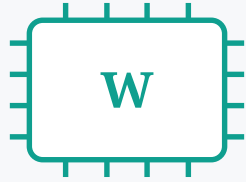
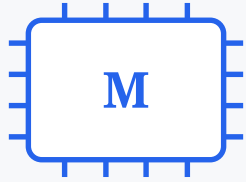
Simulates

same relevant behavior

Inductive

property is self-proving

Certificate Conditions



$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P} \rangle \quad \mathcal{W} = \langle \mathcal{I}', \mathcal{L}', \mathcal{R}', \mathcal{F}', \mathcal{C}', \mathcal{P}' \rangle$

Simulates

same relevant behavior

Inductive

property is self-proving

Reset

$$R\{L \cap L'\} \wedge C \Rightarrow R'\{L \cap L'\} \wedge C'$$

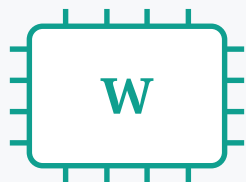
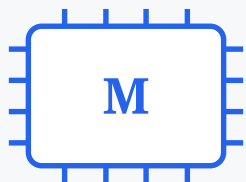
Transition

$$F_{0,1}\{L \cap L'\} \wedge C_0 \wedge C_1 \wedge C'_0 \Rightarrow F'_{0,1}\{L \cap L'\} \wedge C'_1$$

Property

$$(C \wedge C') \Rightarrow (P' \Rightarrow P)$$

Certificate Conditions



$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P} \rangle \quad \mathcal{W} = \langle \mathcal{I}', \mathcal{L}', \mathcal{R}', \mathcal{F}', \mathcal{C}', \mathcal{P}' \rangle$$

Simulates

same relevant behavior

Inductive

property is self-proving

Reset

$$R\{L \cap L'\} \wedge C \Rightarrow R'\{L \cap L'\} \wedge C'$$

Transition

$$F_{0,1}\{L \cap L'\} \wedge C_0 \wedge C_1 \wedge C'_0 \Rightarrow F'_{0,1}\{L \cap L'\} \wedge C'_1$$

Property

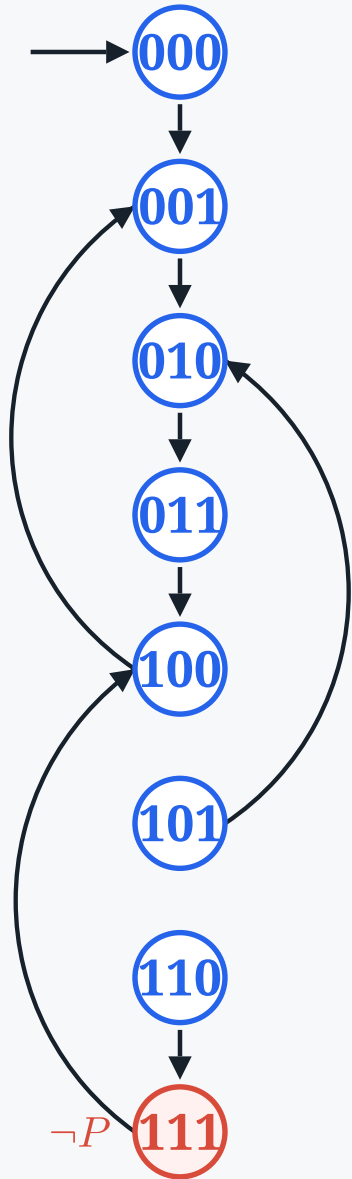
$$(C \wedge C') \Rightarrow (P' \Rightarrow P)$$

Base

$$R'\{L'\} \wedge C' \Rightarrow P'$$

Step

$$P'_0 \wedge F'_{0,1}\{L'\} \wedge C'_0 \wedge C'_1 \Rightarrow P'_1$$



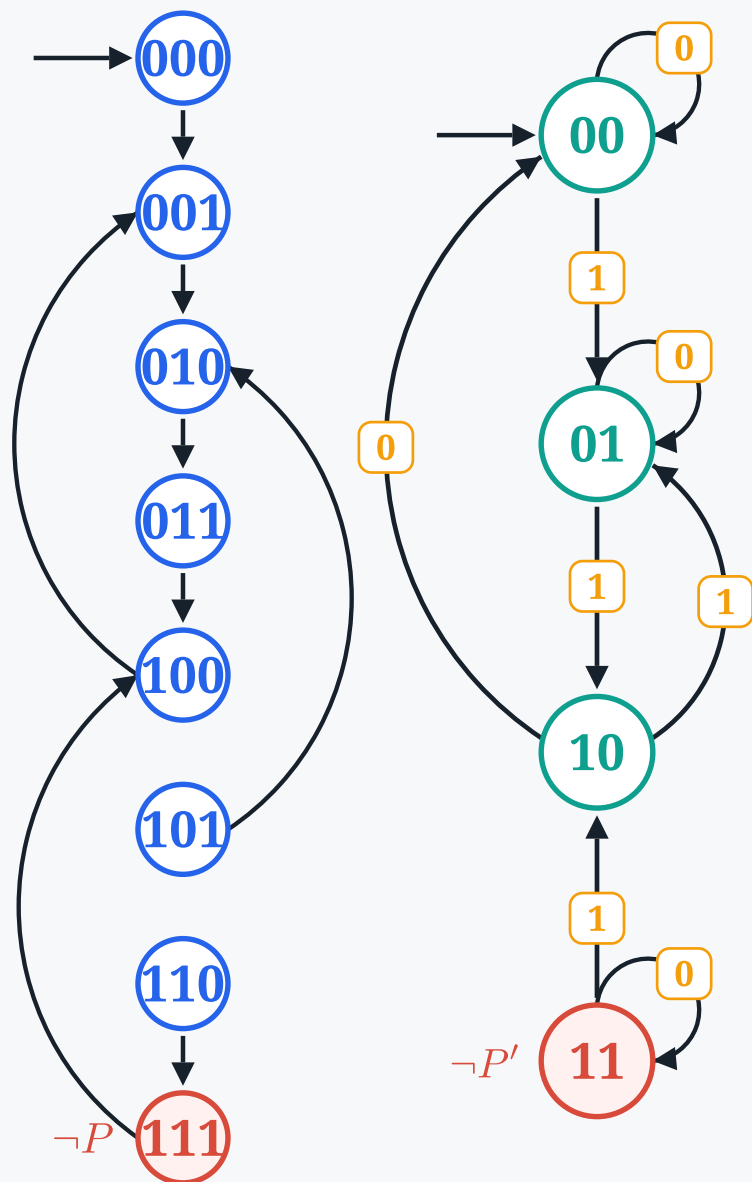
Reset $R\{L \cap L'\} \wedge C \Rightarrow R'\{L \cap L'\} \wedge C'$

Transition $F_{0,1}\{L \cap L'\} \wedge C_0 \wedge C_1 \wedge C'_0 \Rightarrow F'_{0,1}\{L \cap L'\} \wedge C'_1$

Property $(C \wedge C') \Rightarrow (P' \Rightarrow P)$

Base $R'\{L'\} \wedge C' \Rightarrow P'$

Step $P'_0 \wedge F'_{0,1}\{L'\} \wedge C'_0 \wedge C'_1 \Rightarrow P'_1$



Reset

$$R\{L \cap L'\} \wedge C \Rightarrow R'\{L \cap L'\} \wedge C'$$

Transition

$$F_{0,1}\{L \cap L'\} \wedge C_0 \wedge C_1 \wedge C'_0 \Rightarrow F'_{0,1}\{L \cap L'\} \wedge C'_1$$

Property

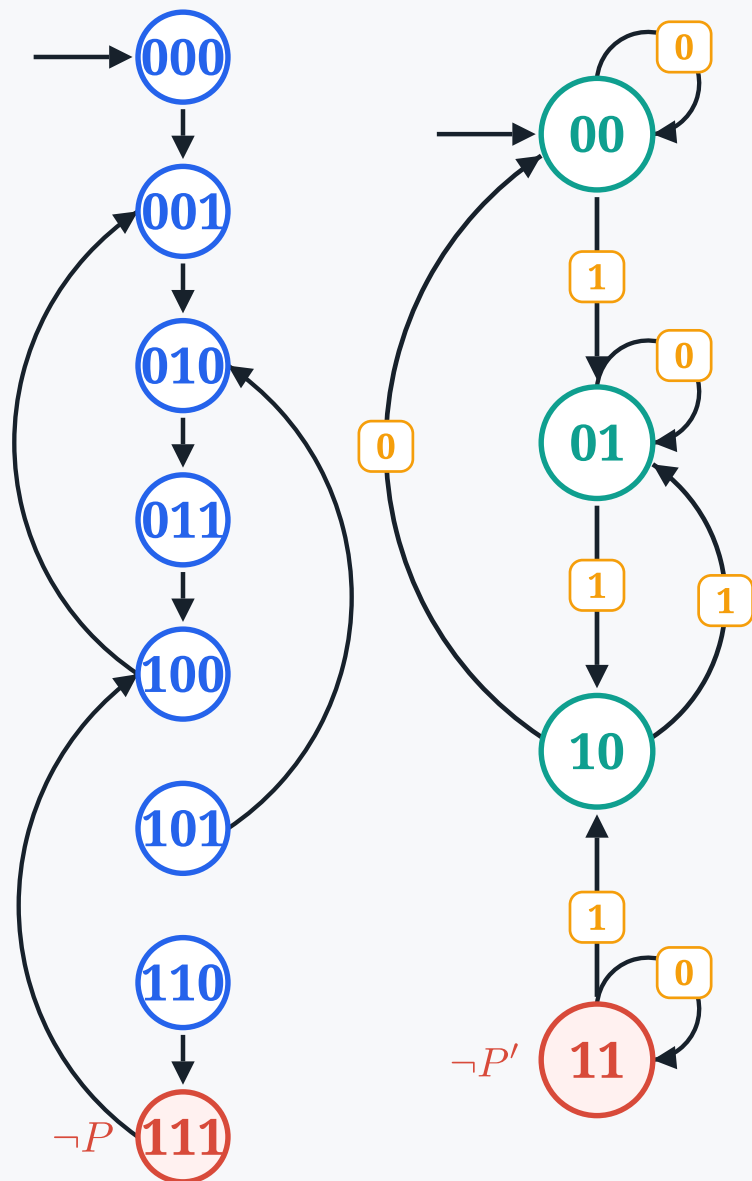
$$(C \wedge C') \Rightarrow (P' \Rightarrow P)$$

Base

$$R'\{L'\} \wedge C' \Rightarrow P'$$

Step

$$P'_0 \wedge F'_{0,1}\{L'\} \wedge C'_0 \wedge C'_1 \Rightarrow P'_1$$



model.aag

```

aag 10 0 3 1 7
2 3
4 13
6 19
20
8 4 3
10 5 2
12 11 9
14 4 2
16 6 4
18 17 15
20 16 14
  
```

witness.aag

```

aag 9 1 2 1 6
2
4 13
6 19
16
8 4 3
10 5 2
12 11 9
14 4 2
16 6 4
18 17 15
  
```

model.btor2

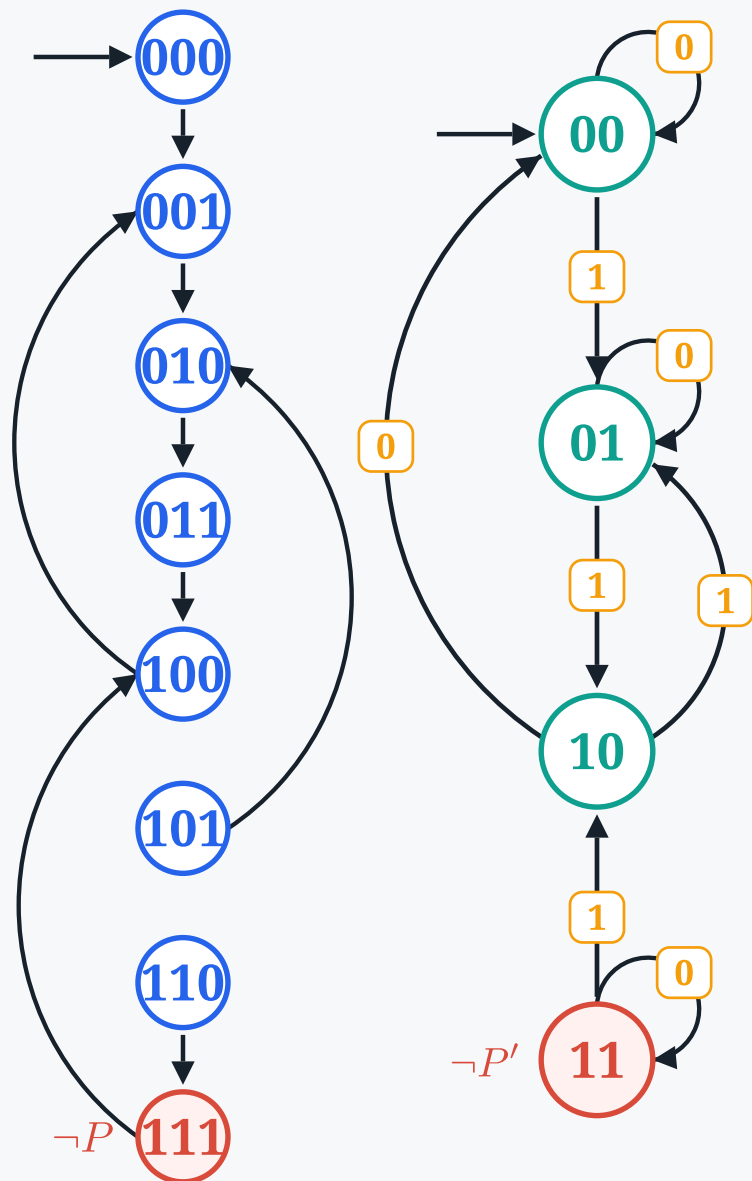
```

1 sort bitvec 1 3
2 sort bitvec 3
3 constd 2 0
4 constd 2 1
5 constd 2 -3
6 constd 2 4
7 constd 2 5
8 constd 2 7
9 state 2
10 init 2 9 3
11 eq 1 9 6
12 eq 1 9 7
13 eq 1 9 8
14 or 1 11 12
15 or 1 13 14
16 ite 2 15 5 4
17 add 2 9 16
18 next 2 9 17
19 bad 13
  
```

witness.btor2

```

1 sort bitvec 1 3
2 sort bitvec 3
3 constd 2 0
4 constd 2 1
5 constd 2 -3
6 constd 2 4
7 constd 2 5
8 constd 2 7
9 state 2
10 init 2 9 3
11 eq 1 9 6
12 eq 1 9 7
13 eq 1 9 8
14 or 1 11 12
15 or 1 13 14
16 ite 2 15 5 4
17 add 2 9 16
18 next 2 9 17
19 constd 2 6
20 eq 1 9 19
21 or 1 13 20
22 bad 21
  
```



model.aag

```
aag 10 0 3 1 7
2 3
4 13
6 19
20
8 4 3
10 5 2
12 11 9
14 4 2
16 6 4
18 17 15
20 16 14
```

witness.aag

```
aag 9 1 2 1 6
2
4 13
6 19
16
8 4 3
10 5 2
12 11 9
14 4 2
16 6 4
18 17 15
```



Certifaiger

model.btor2

```
1 sort bitvec 1 3
2 sort bitvec 1 3
3 constd 2 0
4 constd 2 1
5 constd 2 -3
6 constd 2 4
7 constd 2 5
8 constd 2 7
9 state 2
10 init 2 9 3
11 eq 1 9 6
12 eq 1 9 7
13 eq 1 9 8
14 or 1 11 12
15 or 1 13 14
16 ite 2 15 5 4
17 add 2 9 16
18 next 2 9 17
19 bad 13
```

witness.btor2

```
1 sort bitvec 1 3
2 sort bitvec 1 3
3 constd 2 0
4 constd 2 1
5 constd 2 -3
6 constd 2 4
7 constd 2 5
8 constd 2 7
9 state 2
10 init 2 9 3
11 eq 1 9 6
12 eq 1 9 7
13 eq 1 9 8
14 or 1 11 12
15 or 1 13 14
16 ite 2 15 5 4
17 add 2 9 16
18 next 2 9 17
19 constd 2 6
20 eq 1 9 19
21 or 1 13 20
22 bad 21
```

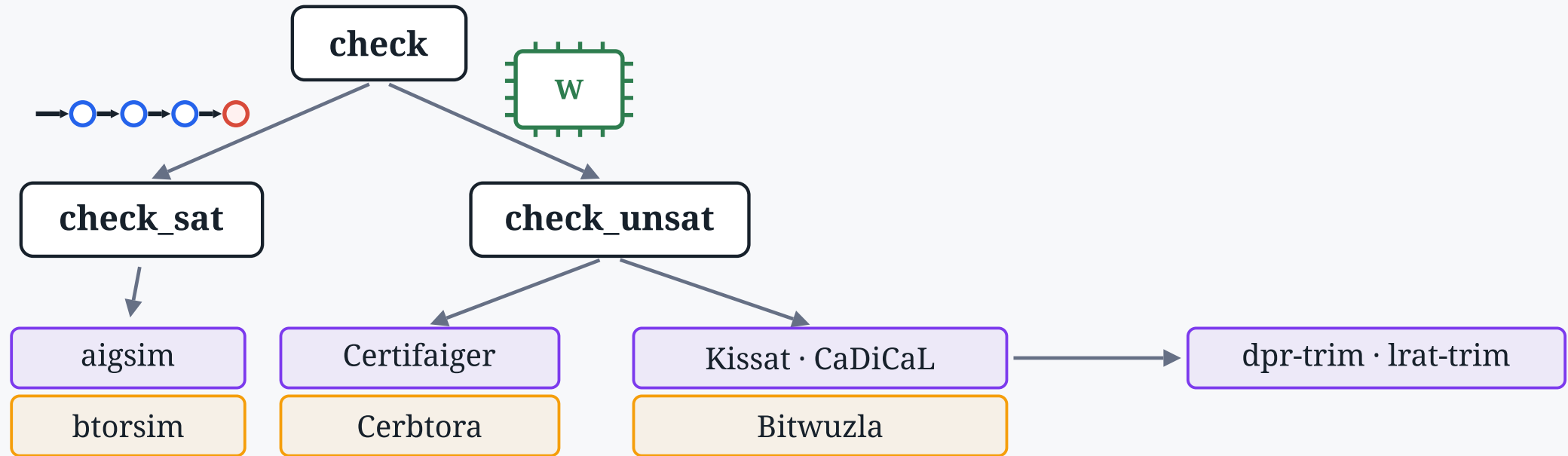


Cerbtora

Tool Integration

bit / AIGER

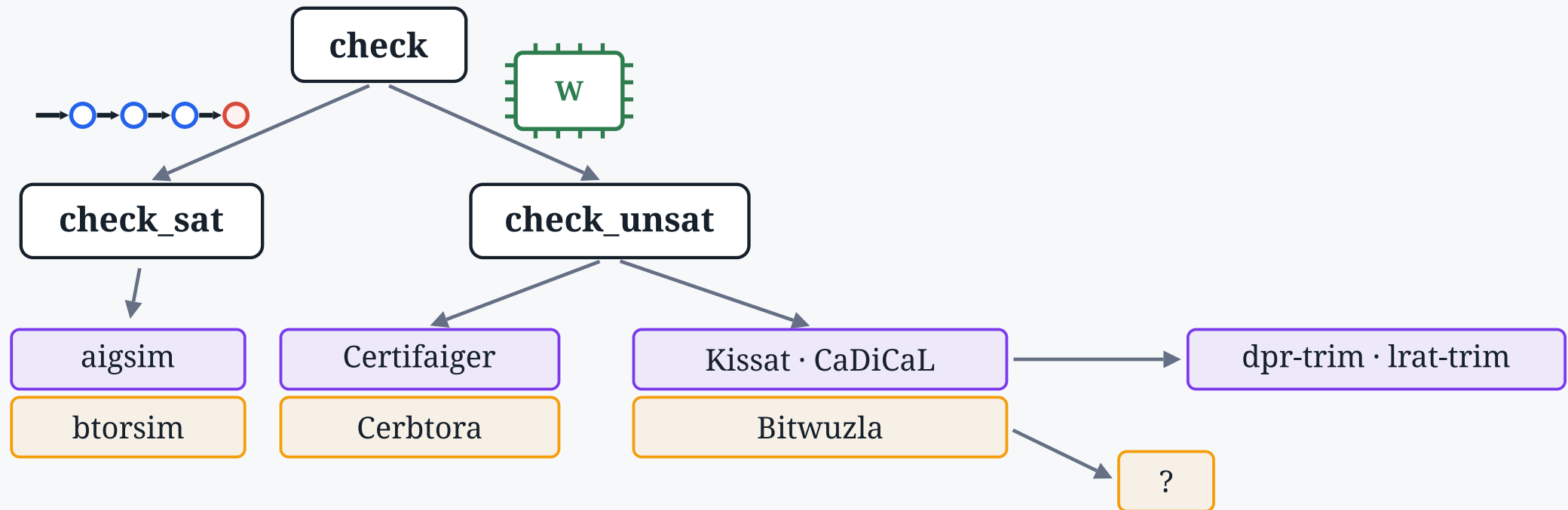
word / BTOR2



Tool Integration

bit / AIGER

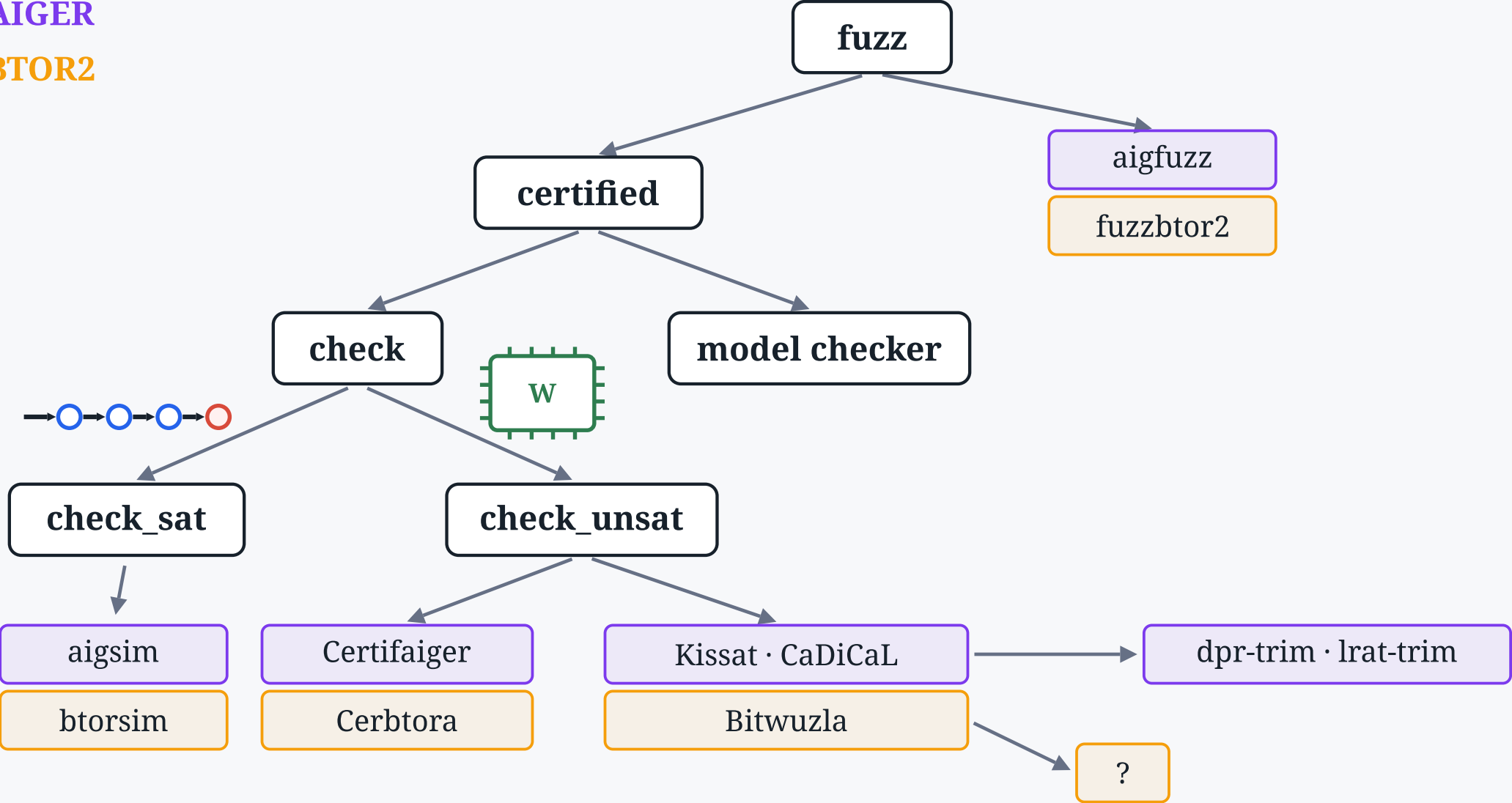
word / BTOR2



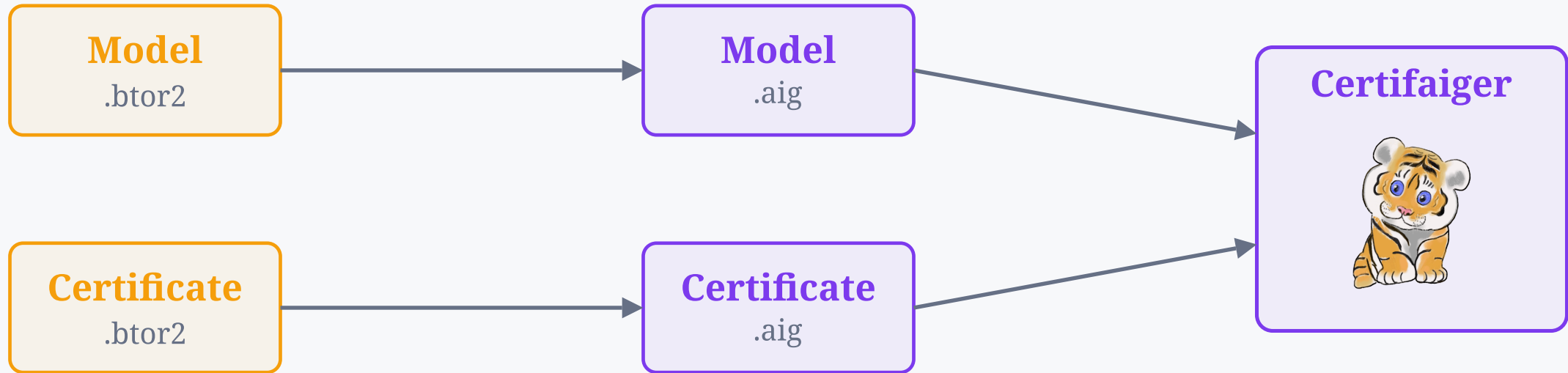
Tool Integration

bit / AIGER

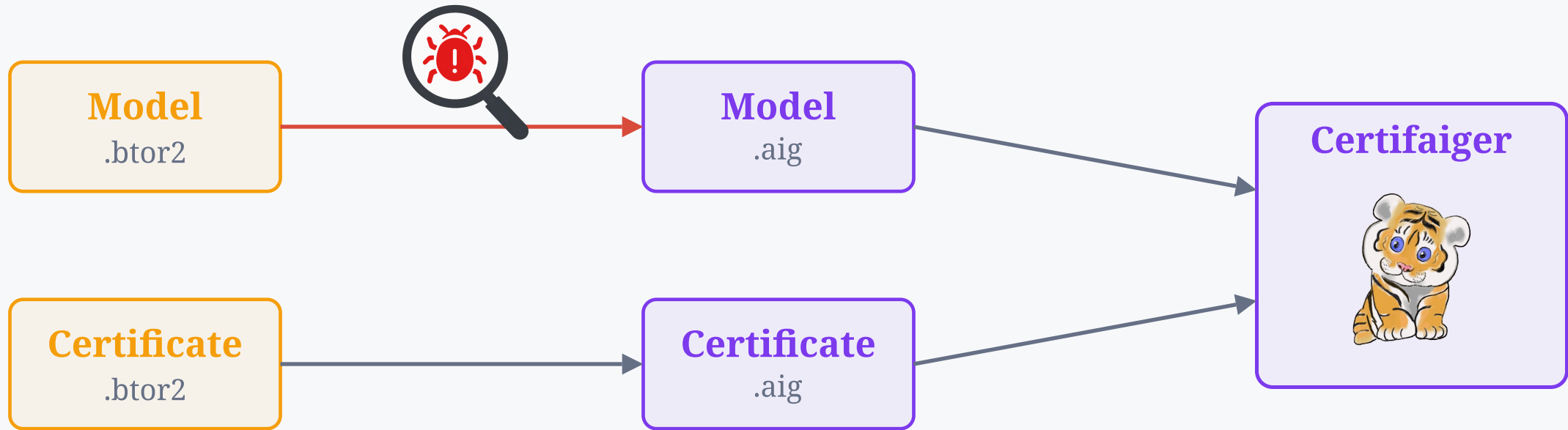
word / BTOR2



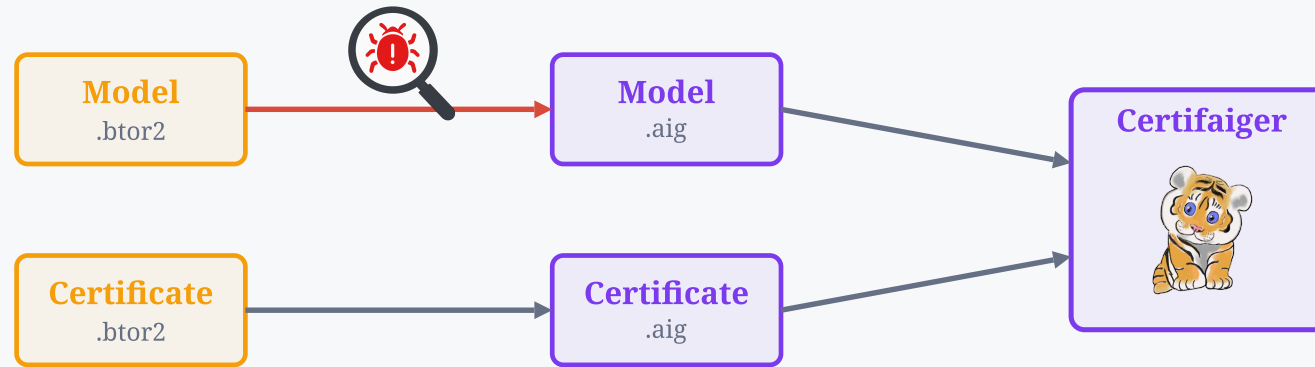
Bit Blasting



Bit Blasting



Bit Blasting



Commits on Nov 26, 2024

fix bitvector rotation for $\text{arg} > \text{width}$



Froleyks committed on Nov 26, 2024

da82ea7



unintialized vectors produce uninitialized latches



Froleyks committed on Nov 26, 2024

4dae1e3



Checker Tuning

1062 HWMCC'24 certificates · 1 h timeout

Configuration	Verified	PAR10	Max
Kissat			
Certifaiger (parallel + UNSAT bias)	1059	180.77	3577.93
parallel	1056	276.35	3155.87
split	1056	277.18	2935.26
merged	1022	1530.25	2469.53
CaDiCaL			
CaDiCaL	1057	189.15	1957.92
Kissat features			
no-sweep	1052	412.88	3529.78
no-congruence	1027	1374.67	3484.84
no-congruence-no-sweep	992	2515.47	3321.83
UNSAT proofs			
CaDiCaL + LRAT	1053	320.44	1097.20
Kissat + DRAT	974	3298.02	3528.02

Formula generation

6 ms

mean · negligible beside SAT

SAT features matter

13.9× PAR10

67 fewer certificates verified

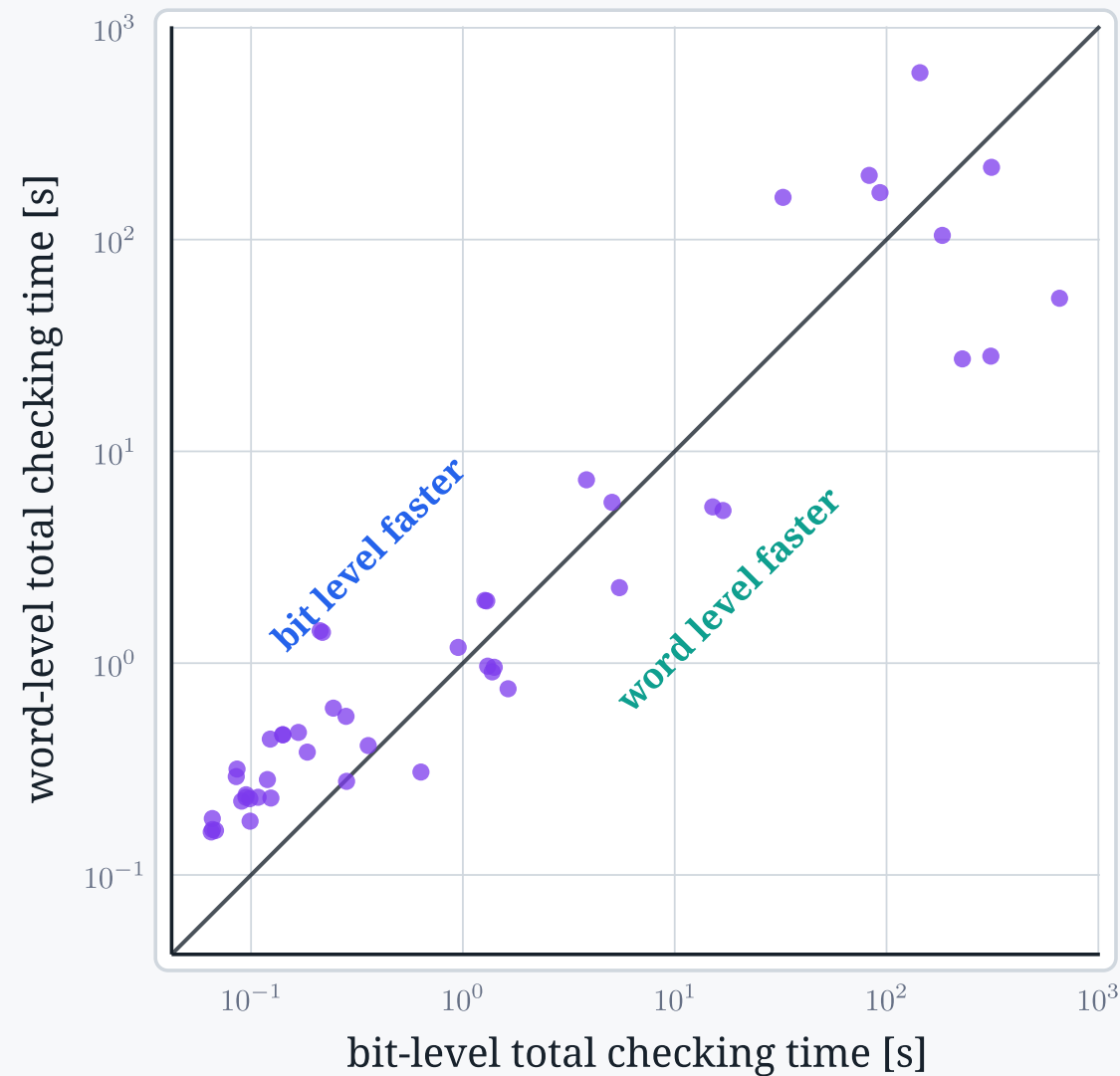
Proof checking overhead

1.69× PAR10

CaDiCaL + LRAT / CaDiCaL

Certificate Checking

47 k-Induction Certificates



Checking time

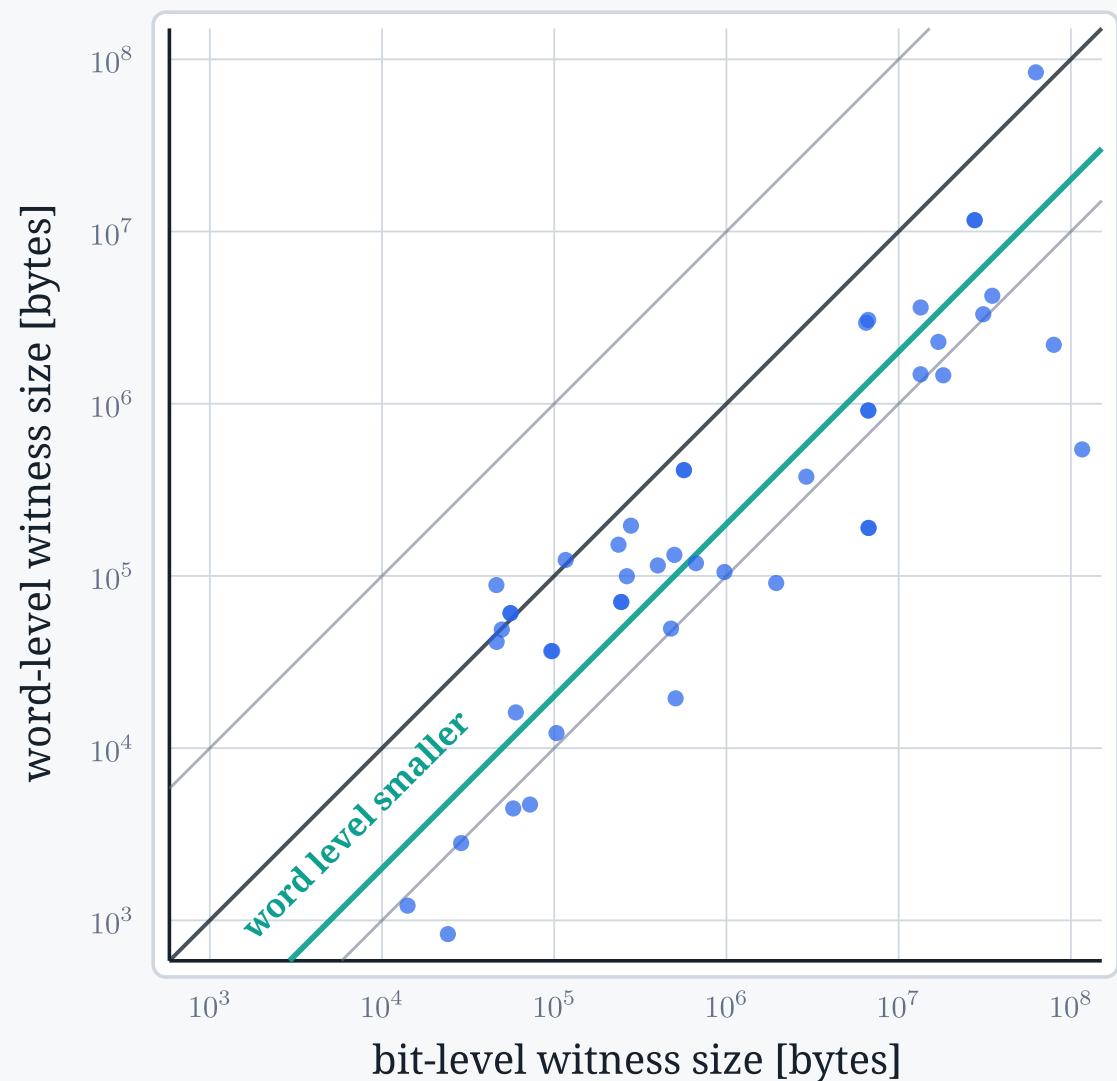
23.1% lower

1616 s word · 2103 s bit

- Prototype certifying word-level model checker
- No congruence closure or sweeping at word level

Certificate Size

47 k-Induction Certificates · .aig vs .btor2



- equal size
- geometric mean
- 10× reference

Geometric mean

5.0× smaller

native word-level witness

Word level smaller

42 / 47

certificates are smaller



Certifaiger

AIGER → SAT



froleyks.de



Cerbtora

BTOR2 → SMT

Witness size

5.0× smaller
word-level · geometric mean

Checking time

23.1% less
word-level · all 47 certificates

CAV · DISTINGUISHED PAPER

Liveness Proofs for Hardware Model Checking

14:50–15:05

Tuesday, 28 July 2026